

Simulation Analysis Techniques for Investment Casting Process of Ni-Base Superalloy Components



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Abstract Development of the Ni-base single crystal (SC) superalloy, named MGA1700 (TMS-1700), has been carried out in a collaborative research between National Institute for Materials Science (NIMS) and Mitsubishi Heavy Industries (MHI). Besides the excellent creep strength and dwell thermo-mechanical fatigue strength, the newly developed alloy for high-efficiency gas turbine blade has no expensive elements such as Rhenium. On the other hand, the mechanical properties or manufacturing yield of SC superalloy blades is quite sensitive to casting process conditions. Therefore, in this study, simulation analysis techniques for various steps of investment casting process, including the solidification process and the mold manufacturing process, have been established to find adequate casting conditions. These analysis techniques can be also applied to other components, such as directionally solidified (DS) rotating blade or conventional casting (CC) rotating blade/stationary vane.

Keywords Investment casting · Casting simulation · Single-crystal superalloy

Introduction

Gas turbine combined cycle (GTCC) power plants have higher-power generation efficiency than the other kinds of power plants operated by fossil fuel. However, there is a strong urge to improve the thermal efficiency further to satisfy the strong demand for a reduction of CO₂ emissions. One of the effective ways to increase the thermal efficiency of GTCC is increasing the gas turbine inlet temperature. Therefore, the gas turbine with turbine inlet temperature of 1700 °C is being

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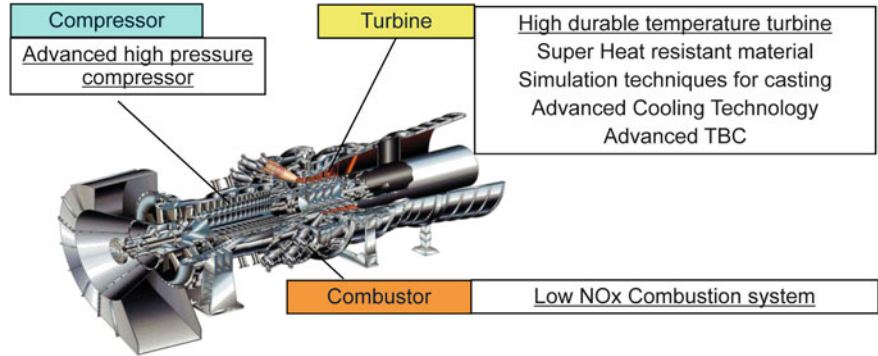
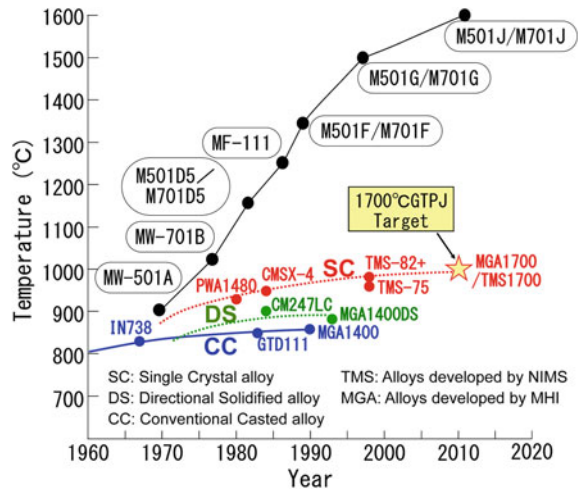


Fig. 1 Technical items for 1700 °C class gas turbine

Fig. 2 Turbine inlet gas temperature of MHI gas turbine and creep durable temperature of superalloys (Creep durable temperature at 138 MPa x 50000 h) [8, 9]



developed in the Japanese national project (Fig. 1) [1, 2]. In this project, some key technologies in the advanced high-pressure compressor, low NOx combustion system, and high durable temperature turbine have been developed. The technologies for high durable temperature turbine include the development of heat resistant materials and the establishment of simulation analysis techniques for investment casting process.

Heat-resistant material is one of the important technologies to realize 1700 °C class gas turbine project. Figure 2 shows the turbine inlet gas temperature of MHI gas turbine and the development trend of superalloys for gas turbine blade. The high-temperature durability of the superalloy has been constantly increased with increasing turbine inlet gas temperature, where SC superalloys have the highest strength due to their high γ' phase volume fraction without grain boundary.

In general, it is difficult to make a SC superalloy having both excellent creep strength and thermo-mechanical fatigue strength at the same time. However, the new SC superalloy MGA 1700 (TMS-1700), developed in a collaborative research between NIMS and MHI, has both of these properties [3, 4]. In addition, MGA1700 has no expensive elements such as Rhenium since they were developed based on the second-generation SC superalloys [5–7].

Simulation analysis for investment casting process is important for SC superalloy blades. Since the mechanical properties or the manufacturing yield are strongly affected by the casting defects and the crystal orientation, simulation analysis is used to find a suitable casting process condition to suppress the casting defects and the unfavorable crystal orientations. The casting defects mainly occur during the solidification process of molten metal, while it is also expected that some defect formations and low-yield rate are depending on the condition of mold manufacturing process. For example, if the injection molding condition of wax pattern model is inadequate, core cracking or short shot may occur. Therefore, in this study, simulation analysis for various steps of investment casting process, not only for solidification process but also for mold manufacturing process, has been established.

Design of Alloy Composition

The aim of development of new SC superalloy is to obtain good high-temperature properties (creep rupture strength, fatigue strength, and oxidation resistance), low manufacturing cost (alloy cost, casting cost, heat treatment cost) and good castabilities. The SC alloys, after the second generation, contain expensive elements, such as Re and Ru, in order to improve the alloy strength remarkably [10]. However, in this project, the total amount of Re and Ru was designed to 3% or less to reduce the alloy cost.

Generally, it is considered that the high-temperature strength of Ni-base superalloy is due to the solid solution strengthening and γ' precipitation strengthening. Therefore, in the superalloy, Al and Ti, etc. are added to promote the γ' phase formation, while W and Mo are added as solid solution strengthening elements. However, the γ' phase forming elements and the solid solution strengthening elements amount should be properly balanced; otherwise, an excess of either of those elements causes phase instability and precipitation of brittle TCP phase such as σ phase.

The composition of the experimental alloys was determined according to the above concept using the materials design flow chart, as shown in Fig. 3. NIMS has been involved in the development of the superalloy for a long time and developed the alloy design program based on the data base of material properties. The alloy design program can estimate metallurgical factors such as γ/γ' phase composition, γ' phase volume fraction, phase stability factor and γ/γ' phase misfit parameter. By applying multiple regression analysis with these factors, the creep strength and

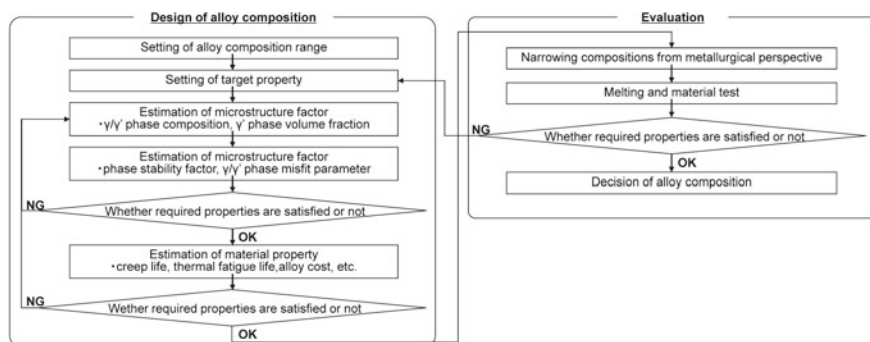


Fig. 3 Alloy design flow of a new superalloy

thermo-mechanical fatigue strength can be estimated. Especially, the creep strength was estimated with good accuracy because of availabilities of a large amount of data of NIMS, but it was not possible for thermo-mechanical fatigue strength due to lack of adequate amount of data. Therefore, the estimation program of thermo-mechanical fatigue strength has been improved by reflecting the tested results in this study.

Mechanical Tests

Microstructures

Based on the results of alloy design program, dozens of superalloys were designed and size of $200 \times 200 \times 20$ mm cast plates was manufactured using ceramic molds. After the heat treatment, round bar and plate shape specimens were machined for mechanical properties measurements. Figure 4 shows the microstructures of a plate specimen of the developed alloy (MGA1700/TMS-1700), which contains no Rhenium. The microstructure was homogenized and no eutectic γ/γ' phase was observed.

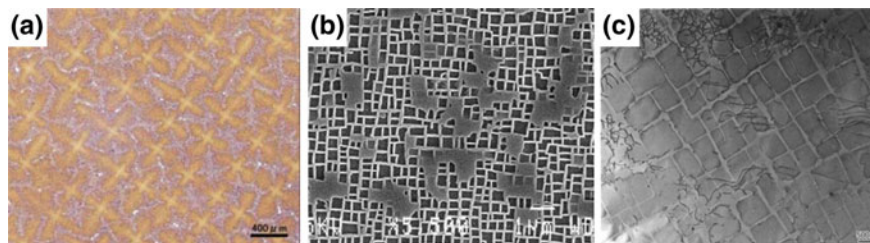


Fig. 4 a Optical micrograph, b SEM image, and c TEM image of MGA1700 (TMS-1700)

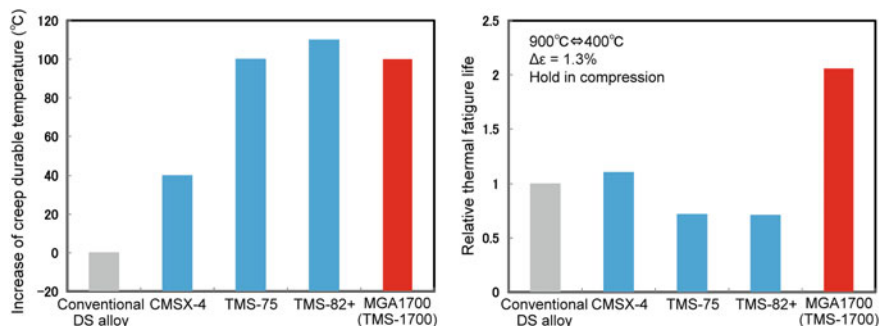


Fig. 5 Increase of creep durable temperature and relative thermal fatigue life

Mechanical Properties

The creep test was carried out at temperature from 900 to 1100 °C. The summary of the creep tests are shown in Fig. 5 in comparison with that of conventional DS superalloy. The creep strength of MGA1700 (TMS-1700) was superior to that of CMSX-4 and identical with the TMS-82 + and TMS-75. The result confirmed that MGA1700 had excellent creep strength.

As considering the actual operating conditions of blades in gas turbine, the thermo-mechanical fatigue test was carried out. The temperature and strain were controlled under out-of-phase condition. The temperature was oscillated in the range of 400–900 °C, and test pieces were held for an hour at 900 °C in compression stress. Since results of fatigue tests tended to vary easily, multiple tests were carried out at the same conditions. The fatigue test results are shown in Fig. 5. As it can be seen, the MGA1700 had excellent fatigue strength, and thus, it satisfied our development objectives.

Moreover, the absence of expensive elements makes the MGA1700 alloy as a potential material for practical application.

Simulation Analysis

Castability of SC Superalloy

For SC superalloy, simulation analysis techniques to evaluate castability are quite important because castability is directly linked to mechanical properties and manufacturing yield. Typical defects in SC alloys are shown in Fig. 6. Since the size of industrial gas turbine blade is quite larger than that of air craft engine, it is very prone to casting defect formation. Therefore, the castability of the developed SC superalloy was evaluated by simulation technique and the best casting conditions were selected to inhibit the casting defects and control the crystal orientation.

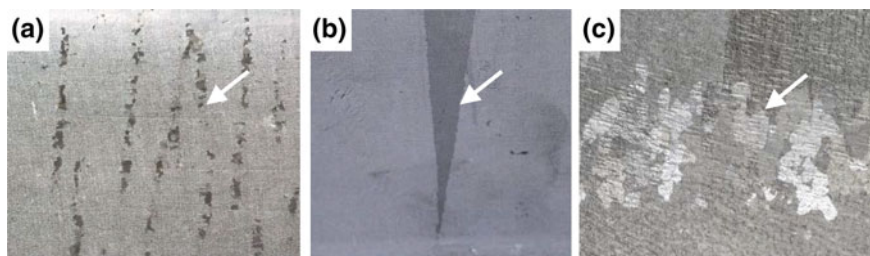


Fig. 6 Typical casting defects in the SC superalloy **a** freckle, **b** stray grain, and **c** re-crystallization

Simulation of Casting Process

Solidification simulation results are used to predict the casting defect generation and the crystal orientation. In order to conduct the simulation more precisely, some casting experiments were carried out. Figure 7 shows the appearance and the schematic diagram of the test furnace and a casting mold. Specimens were cast under various conditions of withdrawing rate and mold heater temperature. After the casting, the casting defects on the cast specimens were checked. The simulation parameters were adjusted based on the comparison of experimental results and previously conducted simulation results.

Figure 8 shows the basic mechanism of freckle generation. Driving force for freckle generation is segregation of compositional elements. In typical Ni-base superalloys, low-density elements (such as Ti or Al) tend to segregate to the liquid and high-density elements (such as W, Re) tend to segregate the solid and concentrate in the dendrite core. These segregations make the liquid lighter in the vicinity of liquid/solid interface and promote convection flow of molten metal. The convection flow, promoted by segregation, causes the freckle generation. On the other hand, the friction force between the liquid and dendrite arms acts as a resistance force of the freckle formation.

To evaluate the freckle formation, the Rayleigh number has been used. The Rayleigh number is the ratio between driving force (liquid density variation) and

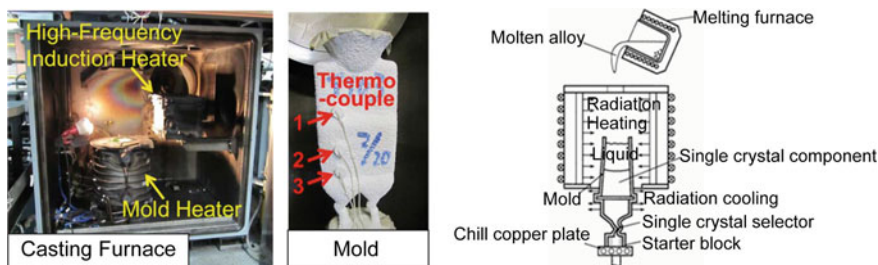


Fig. 7 Appearance and schematic diagram of test furnace and casting mold

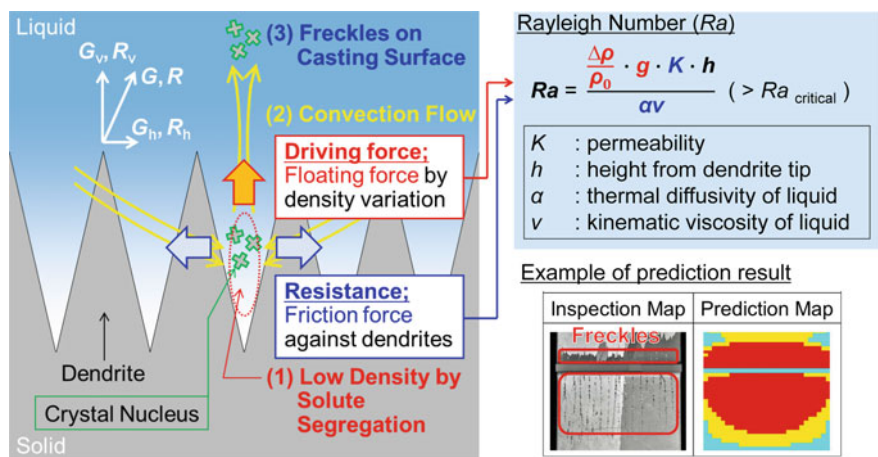


Fig. 8 Basic mechanism and an example of prediction result of freckle generation

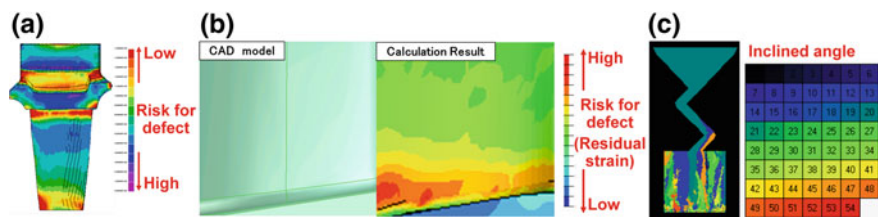


Fig. 9 Predicted results of a stray grain, b re-crystallization, and c crystal orientation

resistance force (permeability of dendrites or liquid properties) [11, 12]. In Fig. 8, the result of experiment and simulation was compared and showed good match between the results.

Figure 9 shows the predicted result of the stray grain and the re-crystallization generation. The stray grain is a crystal grain grows irregularly on the way of continuous crystal growth from the bottom side of the mold to top. The stray grain generation is mainly caused by an imbalance between two solidification parameters, called solidification speed and thermal gradient. Therefore, the stray grain can be predicted using these parameters obtained from the solidification simulation results. Similarly, the re-crystallization occurs in the heat treatment process because of the residual strain generated during the casting process. To predict the re-crystallization, the temperature variation results of solidification simulation were transferred to FEM code, and then the residual strain generated during the solidification and the post-solidification was calculated.

The predicted place of casting defect generation by simulation was in good agreement with the place of casting defect generation on the real blade.

The angle difference between crystal orientation and withdrawing direction in casting process should be minimized to obtain sufficient material strength.

In addition to that, the crystal selection error, developed when more than two crystals from the single-crystal selector enter the component part, should be prevented. To predict the crystal orientation and the crystal selection error generation, the crystal growth simulations by cellular automaton method were carried out. An example of crystal growth simulation result is shown in Fig. 9. These results are used to optimize the mold shapes and the casting conditions.

Simulation of Mold Manufacturing Process

The simulation technique was also used for the mold manufacturing process. The mold manufacturing process includes core/wax injection molding process and de-waxing process. Figure 10 shows the steps involved in investment casting process and the simulation results of filling time in core/wax injection molding process for stationary vanes.

The flow behavior of core/wax and the stress distribution, mainly caused by the thermal expansion, are calculated by the simulations. These results are used for short-shot prediction in core/wax injection molding process or core/mold cracking prediction in wax injection molding process and de-waxing process. By using these simulation techniques, the conditions of mold manufacturing are optimized.

Trial Cast of Real Blade

These analysis techniques were applied to real blade shape and a trial casting of SC blade was carried out. As a result, the large-size IGT blade (over 300 mm) with no crystal defects was obtained. An appearance of trial cast blade is shown in Fig. 11.

Fig. 10 a Various steps of investment casting and examples of simulation results of filling time in injection molding processes of **b** core and **c** wax for stationary vanes

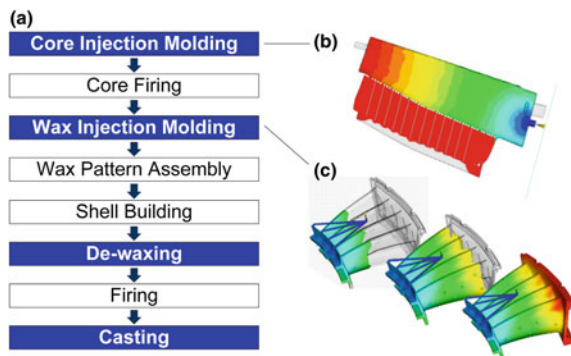


Fig. 11 Appearance of trial cast blade



Conclusions

The development of Ni-base single crystal superalloy was carried out in collaboration research between National Institute for Materials Science (NIMS) and Mitsubishi Heavy Industries (MHI). The developed alloy (MGA1700/TMS-1700) had the excellent creep strength properties and the excellent dwell thermo-mechanical fatigue properties; besides, it also had no expensive elements such as rhenium.

The simulation analysis technique for various steps of investment casting process, including the solidification process, core/wax injection molding process, and de-waxing process, was established to find adequate casting conditions. The established technique was very helpful to predict the casting defects, the crystal orientation, the short shot of core/wax and the core/mold cracking.

A trial casting test was also carried out according to the results obtained from the experiment and simulation and found that the large-size IGT blade was defect-free.

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